THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ATTENDANCE MANAGEMENT SYSTEM USING COMPUTER VISION

UCS 503 Software Engineering Project Report

Mid-Semester Evaluation

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1. PROJECT SELECTION PHASE

1.1 SOFTWARE BID/ FEASIBILITY STUDY

An independent feasibility study for the Attendance Management System using Computer Vision, covering each criterion as outlined in the sample.

Feasibility Dimension	Summary	Key Action/Condition
Technical	**Achievable but requires specialized components.**	Needs **Computer Vision algorithms**, dedicated **biometric devices**, and **trained personnel**.
Economic	**Viable long-term.**	**High upfront investment vs. long-term savings from reduced errors and labour; requires Benefit-Cost Analysis**
Financial	**Budget-dependent.**	Must secure **initial funds** for equipment/licensing and allocate funds for **ongoing maintenance**.
Managerial	**Requires collaboration.**	Need strong **IT-Academic staff coordination** and effective **user training** programs.
Cultural	**Sensitive.**	Must manage **privacy concerns** with biometric data and implement **change management** to ensure user adoption.
Social	**Positive outcome expected.**	Enhances **transparency and accuracy**; requires **communication** to build trust.
Safety	**Crucial.**	Must ensure **data protection**, secure device installation, and prevent misuse of **biometric data**.
Political	**Compliance is key.**	Requires adherence to **institutional/governmental policies** on biometrics and securing **legal approvals**.

1.2 PROJECT OVERVIEW

Problem Statement: Lack of a standardised system to take attendance in lecture halls leads to excessive time consumption, multiple proxy attendances, and inaccuracy in data collected.

Solution: The proposed system is an AI-powered attendance portal designed to make classroom attendance smarter, faster, and more reliable. Instead of relying on outdated manual roll calls or error-prone methods, the solution uses computer vision to recognize students through classroom cameras in real time. By applying advanced face detection and recognition algorithms, the system identifies each student, cross-verifies them with the database, and marks attendance automatically. At the same time, a headcount detection module ensures that the number of people physically present matches the list of registered students. This minimizes errors, prevents proxy attendance, and gives faculty a clear picture of participation. To further strengthen security, the system introduces a biometric validation step at the start and end of each class. Students authenticate themselves using fingerprint or face scans before entering and while leaving, ensuring that no one can misuse the system by leaving midway or asking a friend to sit in their place. This double-layered verification ensures both presence and participation throughout the session.

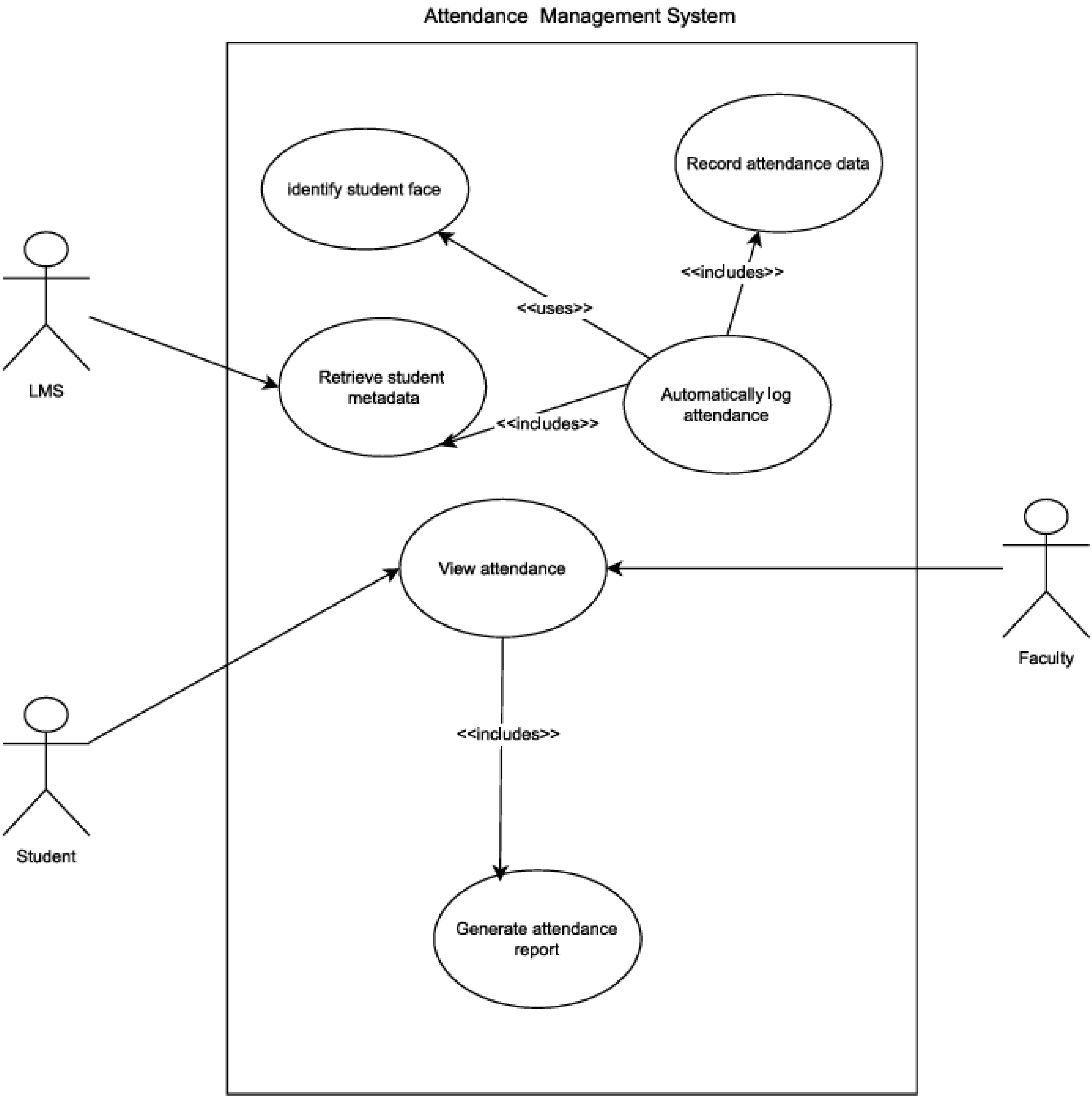
Beyond attendance, the portal can also:

- Generate instant reports for faculty and administrators.
- Analyse attendance trends to identify irregular students early.
- Integrate with Learning Management Systems (LMS) to keep records centralized and accessible.

2. ANALYSIS PHASE

2.1 USE CASES

2.1.1 Use Case Diagram

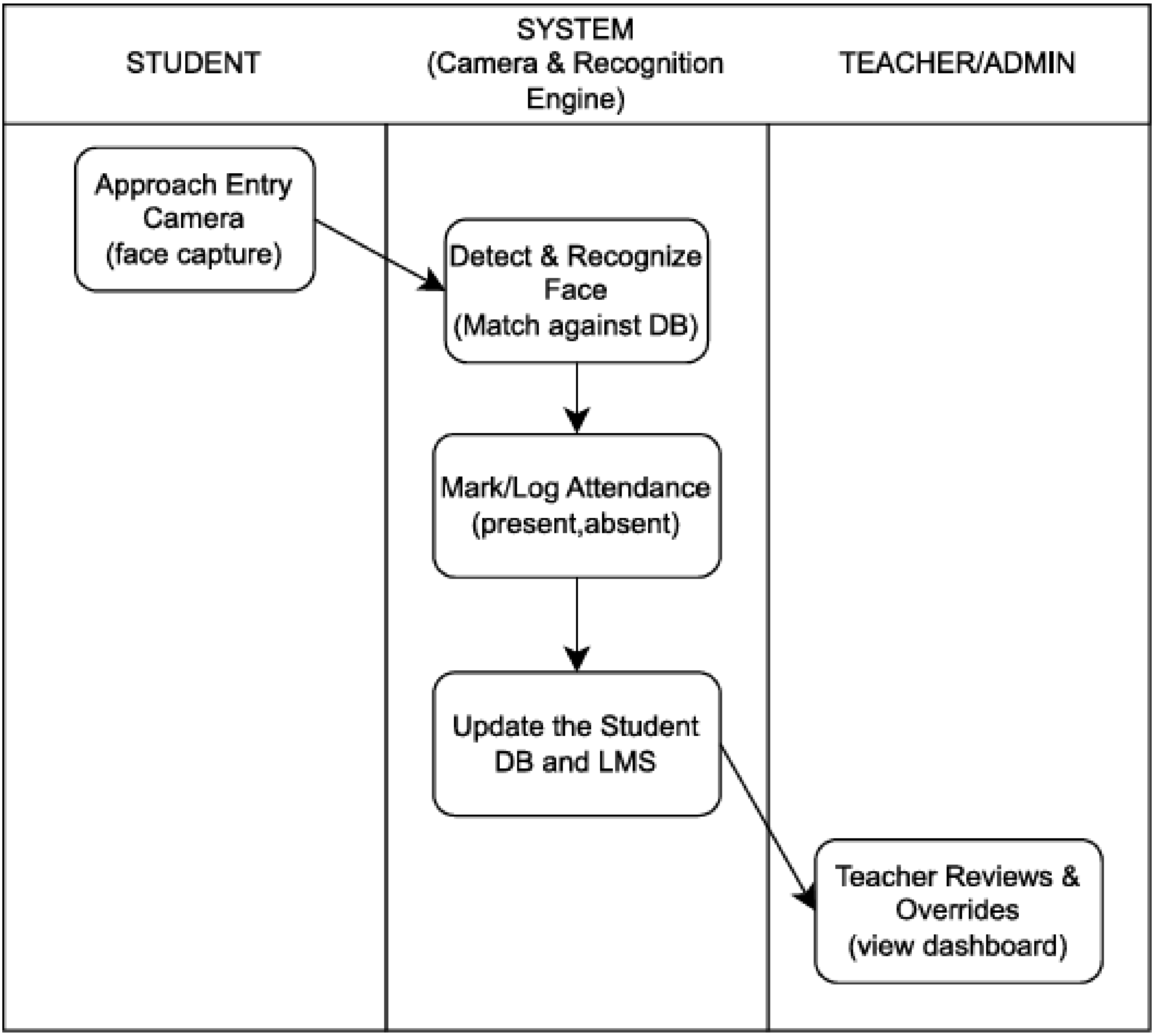


2.1.2 Use Case Template

Field	Content for Attendance Management System
1. Use Case Title	**Enrol Student Face Profile**
2. Abbreviated Title	Enrol Face
3. Use Case Id	UC-002
4. Actors	Administrator, Faculty
5. Description	Allows an authorized user to capture, process, and store the unique facial features (embeddings) of a student. This profile is linked to the student's metadata (ID, Name) to enable future attendance recognition.
5.1. Pre Conditions	1. The Administrator/Faculty must be **logged into the system**.
	2. The student's **metadata (ID, Name)** must already exist in the database (or be retrieved from the LMS).
	3. The **camera device** used for enrolment must be connected and calibrated.
5.2. Task Sequence	1. Admin navigates to the 'Enrolment' module.
	2. Admin enters the Student ID and selects "Begin Enrolment."
	3. The system retrieves the student's metadata from the database/LMS.
	4. The system activates the camera and displays a live feed to the Admin/Student.

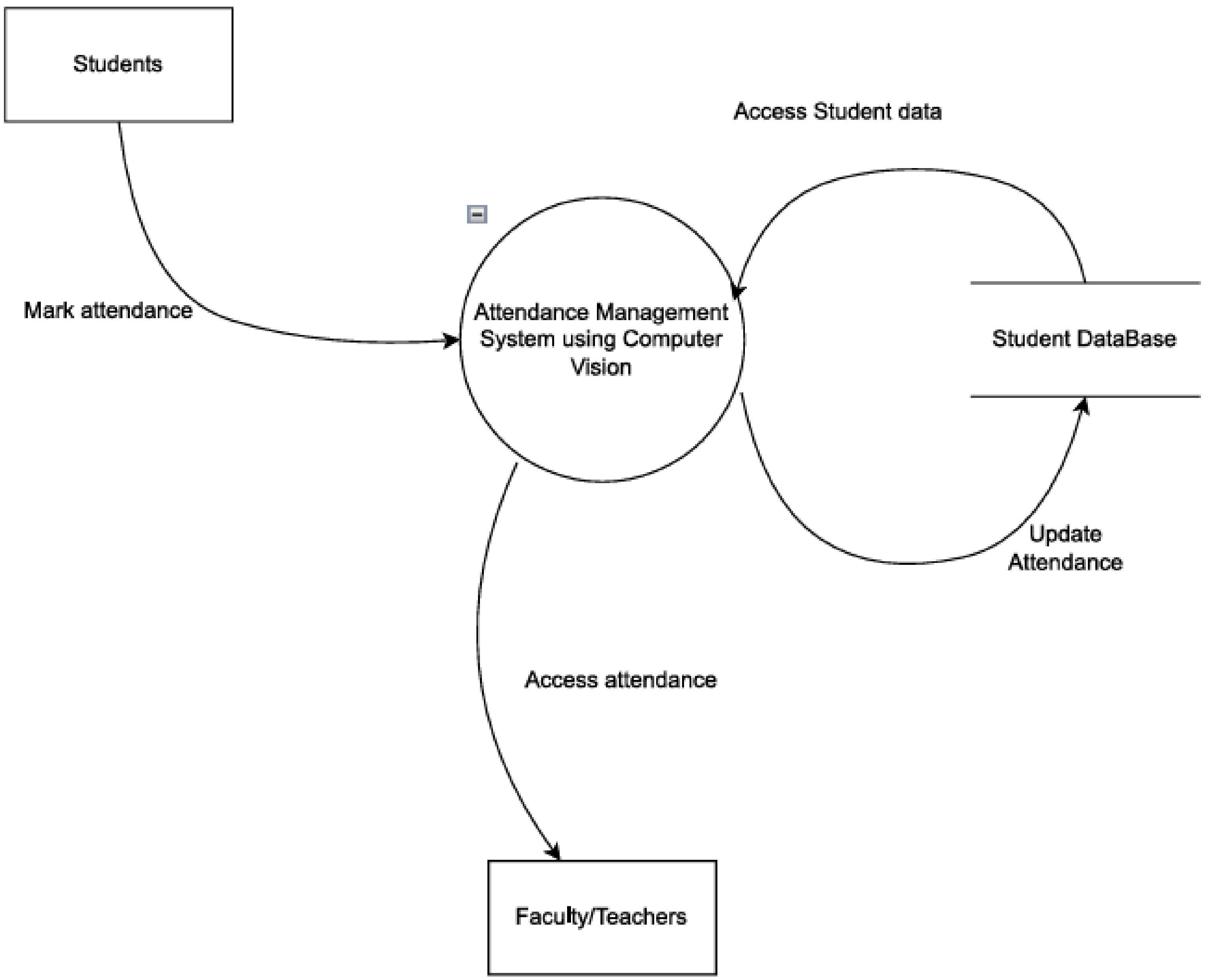
	5. The system captures **multiple images** of the student's face from various angles.
	6. The **Computer Vision Module** processes the images to extract a unique **facial embedding (vector)**.
	7. The system stores the facial embedding vector securely, linked to the Student ID.
	8. The system displays a confirmation message: "Enrolment Complete."
5.3. Post Conditions	1. A unique **Facial Embedding Record** is successfully created in the Attendance DB.
	2. The student is now recognizable by the **Automatically Log Attendance** use case (UC-001).
	3. The Admin/Faculty can exit the module or enrol another student.
6. Modification History:	Date 27-Oct-2025
7. Author:	Edwin, Dhanveer, Mohak

2.2 ACTIVITY DIAGRAM AND SWIMLANE DIAGRAMS

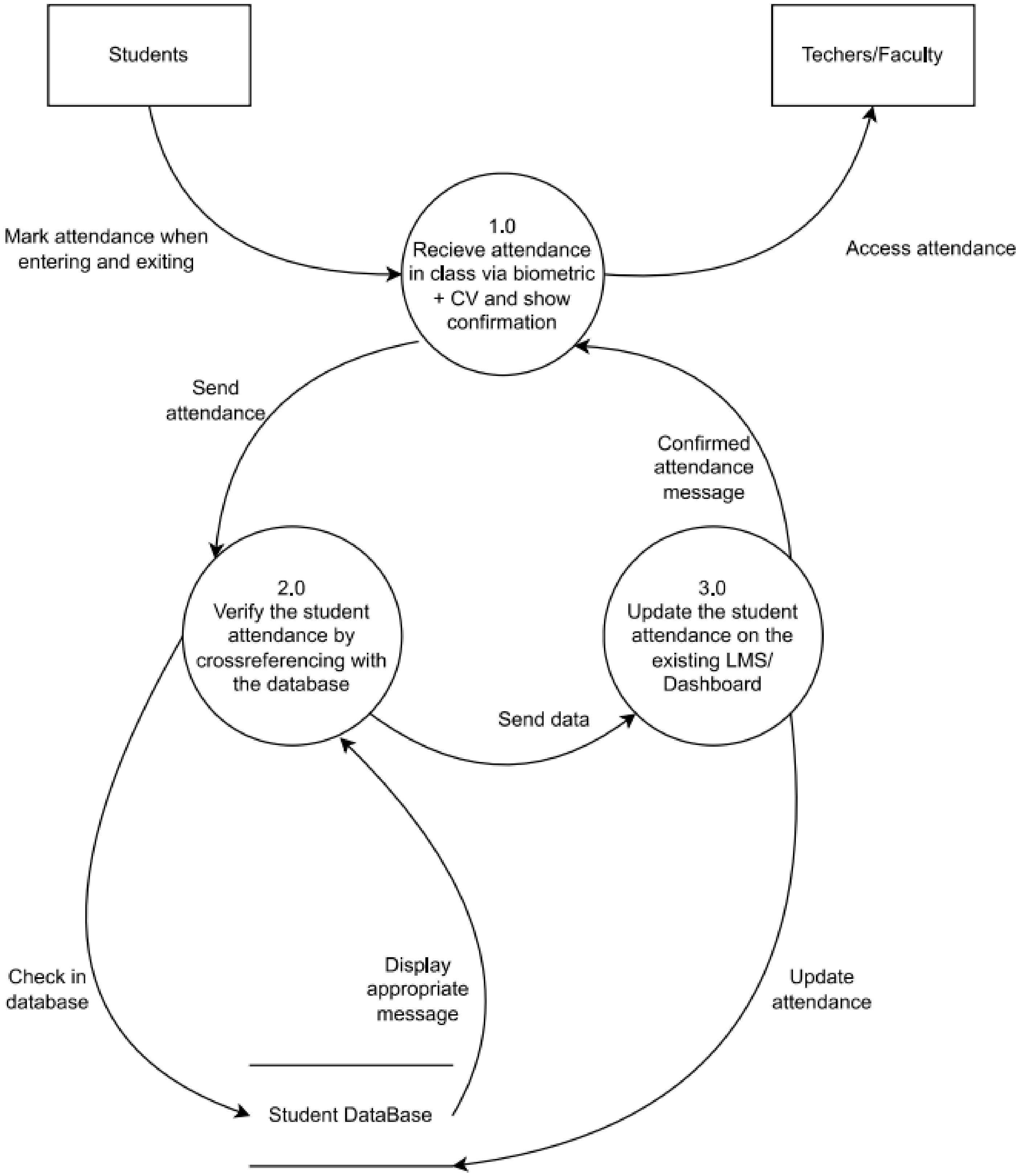


2.3 DATA FLOW DIAGRAMS

2.3.1 DFD Level 0



2.3.2 DFD Level 1



2.4 SOFTWARE REQUIREMENT SPECIFICATION (IEEE FORMAT)

Software Requirements Specification (SRS)
Project Title: Face Recognition Attendance Portal
Prepared by: Team
Version: 1.0
Date: 2025-10-27

1. Introduction

1.1 Purpose

This SRS describes the requirements for the Face Recognition Attendance Portal — a computer vision-driven system that scans faces at classroom entrances, marks attendance automatically, and continuously monitors students inside the classroom to ensure they remain present during the lecture. The document is intended for faculty reviewers, the development team, testers, and deployment engineers.

1.2 Scope

The system will provide:

- Real-time face detection and recognition at classroom entry points to mark attendance.
- Continuous monitoring via classroom cameras to detect students leaving the classroom after being marked present.
- Integration with institutional student database for authentication, student metadata, and attendance record updates.
- Admin dashboard for teachers/administrators to view attendance reports, override entries, and generate analytics.
- Alerting mechanism (optional) to notify faculty if a student exits without permission.

Boundaries:

- The system tracks presence using face recognition only; it does not rely on GPS or manual sign-ins aside from overrides.
- The system does not perform biometric authentication for financial or legal transactions; it is strictly for attendance monitoring.

1.3 Definitions, Acronyms, Abbreviations

CV: Computer Vision
SRS: Software Requirements Specification
API: Application Programming Interface
DBMS: Database Management System
FPS: Frames Per Second
GDPR: General Data Protection Regulation (where applicable)

2. Overall Description

2.1 Product Perspective

The portal is a modular system with the following high-level components:

1. Edge Camera Module — cameras (entry and classroom) running lightweight detection to gather face captures.
2. Recognition Engine — server-side CV models that extract face embeddings and match against the student database.
3. Attendance Service — business logic to decide 'present/absent/left' and time-stamping.
4. User Interface (Web Dashboard) — for teachers/admins to view and manage attendance and settings.
5. Database — persistent store for student profiles, enrollment, attendance logs, and system configuration.
6. Notification Service — optional alerts via email/SMS/in-app.

3. Functional Requirements (System Features)

FR-1: Student Registration & Enrolment — The system must allow admin to register student profiles with face images and metadata. Each student profile shall have a unique Student ID, name, enrolled courses, and one or more reference face captures.
FR-2: Real-time Entry Recognition — When a student passes the entry camera, the system will detect a face, extract embedding, and match with the student DB within 2 seconds. If match confidence ≥ threshold, mark student present with timestamp.
FR-3: Continuous Presence Monitoring — Classroom camera(s) will monitor students during session. If a student leaves the room after being marked present and does not return within a configurable grace period (default 2 minutes), mark as 'left' and update attendance status.
FR-4: Manual Override & Verification — Teachers/Admins must be able to manually mark attendance, correct mismatches, or flag uncertain recognition results for review.
FR-5: Dashboard & Reports — Web dashboard for teachers to view class-wise attendance, daily/weekly reports, export CSV/PDF, and visualize absence patterns.
FR-6: Alerts & Notifications — Optional emails/SMS/in-app alerts to notify faculty when a student exits unexpectedly or when camera/recognition system fails.
FR-7: Audit Logs — All automatic and manual actions must be logged with actor, timestamp, and reason for audit compliance.
FR-8: Integration APIs — Provide REST API endpoints to query attendance, push student data, and fetch logs. Authentication via token-based mechanism (e.g., JWT).

4. External Interface Requirements

4.1 User Interfaces

Responsive Web UI: Login, Dashboard, Class Session view (live feed + presence markers), Student Profile management, Reports, Settings.

Use modern UI frameworks (React/Vue) and ensure accessibility (keyboard navigation, readable contrast).

4.2 Hardware Interfaces

Entry Camera: 1080p recommended, 30 FPS.
Classroom Camera: 1080p, wide-angle to cover seats.
Edge device or NVR capable of streaming to Recognition Engine.
Optional: GPU-enabled server for model inference.

4.3 Software Interfaces

Recognition Engine: expose gRPC/REST interface to receive image frames and return recognition results.
Database: PostgreSQL/MySQL.
Authentication: OAuth2/JWT.

4.4 Communications Interfaces

Use TLS encrypted channels for API calls (HTTPS).
Use MQTT or WebSocket for real-time notifications/alerts to dashboard.

5. Non-functional Requirements

- **Performance:** System shall process an entry recognition in ≤ 2 seconds with average latency; classroom monitoring should update presence state within 5 seconds.
- **Scalability:** Support at least 10 concurrent classrooms per server; support horizontal scaling for larger deployments.
- **Security:** All personal data encrypted at rest and in transit. Role-based access control (RBAC) for admin, teacher, and auditor roles. Secure key management for API tokens.
- **Privacy:** Retention policy configurable; default: raw images deleted after 7 days, embeddings retained for 1 year. Consent required for biometric data storage; follow local laws.

- **Reliability & Availability:** Target 99% uptime for core services during academic hours. Automatic health checks and failover provisioning.
- **Usability:** Dashboard should be intuitive; completion of common tasks (view class roster, export report) in ≤ 3 clicks.
- **Maintainability:** Modular codebase with automated tests and CI/CD pipelines. Clear API contracts and versioning.

6. Data Requirements and Data Flow (summary)

- Key entities: Student, Course, ClassSession, AttendanceRecord,
- CameraDevice, AuditLog.
- Referential integrity must be enforced. AttendanceRecord stores StudentID, SessionID, status (present/absent/left), timestamps, confidence score, and source (auto/manual).

7. System Architecture & Modules

High-level modules:

- Edge Capture Module
- Recognition Engine (face embedding + matcher)
- Attendance Service (rules engine)
- Web Dashboard & API
- Database & Storage
- Notification Service
- Monitoring & Logging

8. Acceptance Criteria and Test Overview

Acceptance tests will include: unit tests for recognition pipeline, integration tests for API and DB, end-to-end trials in a mock classroom for 100 students, performance/load tests, privacy compliance checks, and user-acceptance testing with faculty.

9. Deployment and Operations

Deployment options: On-premise server with GPU or cloud deployment.
Containerized microservices (Docker + Kubernetes).
Backup schedules, monitoring (Prometheus + Grafana), and regular retraining pipelines for recognition models.

10. Risks and Mitigations

Privacy Backlash: Mitigation — Consent forms, opt-out mechanisms, minimization of stored data, and legal review.
False Positives/Negatives: Mitigation — Allow manual overrides, maintain confidence thresholds, and retrain models regularly.
Hardware Failure: Mitigation — Redundant cameras, periodic health checks, and quick-replacement procedures.
Budget Overruns: Mitigation — Phased rollout and open-source model choices to reduce licensing costs.

2.5 USER STORIES AND USER CARDS

1. Student User Stories

US-01: As a student, I want my face to be recognized automatically inside the classroom so that attendance is marked without roll calls.

- **Acceptance Criteria:** Recognition within 2 seconds; attendance marked as present.
- **Priority:** High | **Story Points:** 8

US-02: As a student, I want to view my attendance report online so that I can track my attendance.

- **Acceptance Criteria:** Dashboard displays attendance percentage and date-wise record.
- **Priority:** Medium | **Story Points:** 3

2. Faculty / Teacher User Stories

US-03: As a teacher, I want to see a live list of students detected in class so that I can monitor attendance in real time.

- **Acceptance Criteria:** Dashboard updates automatically with presence status.
- **Priority:** High | **Story Points:** 5

US-04: As a teacher, I want to manually correct or override attendance so that recognition errors can be fixed.

- **Acceptance Criteria:** Manual override possible with reason logged.
- **Priority:** High | **Story Points:** 4

US-05: As a teacher, I want to generate and export class attendance reports so that I can submit official records.

- **Acceptance Criteria:** Reports downloadable in PDF/Excel format.
- **Priority:** Medium | **Story Points:** 4

3. Administrator User Stories

US-06: As an admin, I want to configure cameras and integrate student data from the LMS so that classrooms are automatically managed.

- **Acceptance Criteria:** Cameras assigned to classrooms; LMS data synced via API.
- **Priority:** High | **Story Points:** 6

US-07: As an admin, I want to monitor system health and camera status so that failures can be addressed quickly.

- **Acceptance Criteria:** Dashboard shows live device and service status.
- **Priority:** Medium | **Story Points:** 4

4. System / Technical Stories

US-08: As a system, I must recognize faces in real time with ≥95% accuracy so that attendance remains reliable.

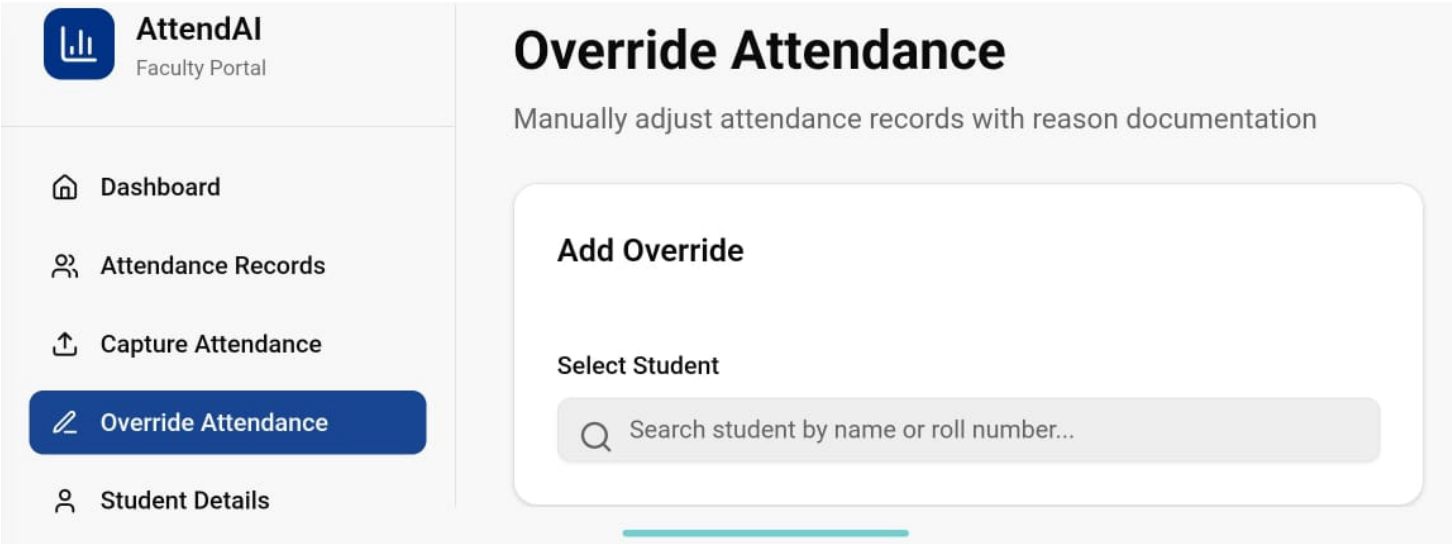
- **Acceptance Criteria:** Recognition accuracy ≥95% under standard lighting.
- **Priority:** High | **Story Points:** 8

US-09: As a system, I must log all recognition events securely with timestamps and confidence scores for audit.

- **Acceptance Criteria:** Logs stored with student ID, confidence, and time.
- **Priority:** High | **Story Points:** 5

US-10: As a system, I must encrypt biometric data and use secure APIs so that user privacy and compliance are maintained.

- **Acceptance Criteria:** AES-256 encryption at rest; HTTPS and token-based authentication in transit.
- **Priority:** High | **Story Points:** 5



USER CARDS

User Card ID	User Story	Acceptance Criteria	Priority	Story Points
UC-01: Automated Attendance Capture	As a **teacher**, I want cameras inside classrooms to automatically capture student faces during class so that attendance is recorded without manual effort.	• Cameras detect and identify students during class. • Attendance is logged automatically after the session. • Face recognition accuracy ≥ 90% under normal lighting.	High	8
UC-02: LMS Integration	As an **admin**, I want the system to fetch student details directly from the LMS so that I don't need to manually upload or update records.	• Student data synced automatically from LMS. • Updates in LMS reflect in real time. • No manual entry required.	High	5
UC-03: Attendance Dashboard	As a **teacher**, I want to view a real-time attendance dashboard so that I can monitor which students are present or absent.	• Dashboard auto-updates per session. • Displays attendance percentage • Filter by date, class, or subject.	Medium	5
UC-04: Admin Reports	As an **admin**, I want to generate attendance summaries for each class so that I can analyse overall attendance trends.	• Downloadable reports (PDF/CSV). • Filters for department, faculty, and date. • Logs unrecognized faces and errors.	Medium	5

THANK YOU